Light

GCSE Computer Science – Embedded Systems (Set 1)

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GCSE • AQA 8525 • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (AQA 8525)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

What best defines an embedded system?

- ○ **A.** A computer system built into a larger device to perform dedicated functions
- ○ **B.** Any computer that connects to the internet
- ○ **C.** A computer that can be carried in a bag
- ○ **D.** A computer used only for gaming

Q2 (1 mark)

Which of the following is a non-embedded (general-purpose) computer system?

- ○ **A.** A car engine management unit
- ○ **B.** A digital thermostat
- ○ **C.** A laptop computer
- ○ **D.** A traffic light controller

Q3 (1 mark)

Which household appliance typically contains an embedded system?

- ○ **A.** A wooden bookshelf
- ○ **B.** A glass window
- ○ **C.** A programmable dishwasher
- ○ **D.** A manual tin opener

Q4 (1 mark)

What is typically true about software on an embedded system?

- ○ **A.** Users can install any application from an app store
- ○ **B.** Embedded systems run full desktop operating systems like Windows
- ○ **C.** The software must be reinstalled every time the device is used
- ○ **D.** The software is pre-programmed for specific tasks and cannot be changed by the user

Q5 (1 mark)

Why do many embedded systems have simple or no user interface?

- ○ **A.** Because they perform dedicated automatic tasks requiring minimal human interaction
- ○ **B.** Because user interfaces are too expensive for any device
- ○ **C.** Because embedded systems cannot display any information
- ○ **D.** Because all users prefer devices without interfaces

Q6 (1 mark)

Which car component contains an embedded system?

- ○ **A.** The windscreen glass
- ○ **B.** The leather seats
- ○ **C.** The rubber tyres
- ○ **D.** The anti-lock braking system (ABS)

Q7 (1 mark)

Why is low power consumption important for many embedded systems?

- ○ **A.** Low power consumption makes devices more colourful
- ○ **B.** Many embedded systems run on batteries or in locations where power efficiency is essential

- ○ **C.** All computers require exactly the same amount of power
- ○ **D.** Power consumption only matters for desktop computers

Q8 (1 mark)

What distinguishes a general-purpose computer from an embedded system?

- ○ **A.** General-purpose computers can run various user-chosen applications for different tasks
- ○ **B.** General-purpose computers are always larger
- ○ **C.** Embedded systems are always faster
- ○ **D.** General-purpose computers never connect to networks

Q9 (1 mark)

Which consumer electronic device contains an embedded system?

- ○ **A.** A paper book
- ○ **B.** An acoustic guitar
- ○ **C.** A wooden picture frame
- ○ **D.** A smart television

Q10 (1 mark)

Why are embedded systems often considered more reliable for their specific tasks?

- ○ **A.** They are made from more expensive materials
- ○ **B.** Reliability is not affected by system design
- ○ **C.** They have fewer components and simpler software focused on one purpose
- ○ **D.** They run more programs simultaneously

Q11 (1 mark)

What happens when an embedded system is powered on?

- ○ **A.** It begins performing its dedicated function almost immediately
- ○ **B.** It displays a login screen for user authentication
- ○ **C.** It takes several minutes to load an operating system
- ○ **D.** It asks the user which program to run

Q12 (1 mark)

A fitness tracker is an example of which type of system?

- ○ **A.** A general-purpose computer
- ○ **B.** A mainframe computer
- ○ **C.** An embedded system
- ○ **D.** A supercomputer

Q13 (2 marks)

What does it mean for an embedded system to have 'real-time' requirements?

- ○ **A.** The system must always display the current time
- ○ **B.** Real-time means the system operates in slow motion
- ○ **C.** The system only operates during daytime hours
- ○ **D.** The system must respond to inputs within strict time limits to function correctly

Q14 (2 marks)

A school is choosing between a tablet computer and a dedicated e-reader for student reading. Compare these in terms of embedded versus general-purpose characteristics.

- ○ **A.** Both are identical embedded systems
- ○ **B.** The tablet is an embedded system because it is portable
- ○ **C.** The e-reader is more embedded (dedicated to reading with simpler software), while the tablet is more general-purpose (runs various apps for different tasks)
- ○ **D.** Neither device contains any computer system

Q15 (2 marks)

Why is it easier to thoroughly test an embedded system compared to a general-purpose computer?

- ○ **A.** Embedded systems have limited, defined functions, so all possible states and inputs can be systematically tested

- B. Embedded systems are never tested
- C. General-purpose computers have fewer features to test
- D. Testing methods are identical for all computer systems

Q16 (2 marks)

How do embedded systems typically interact with the physical world?

- A. Embedded systems cannot interact with anything physical
- B. Only through keyboard and mouse input
- C. Through sensors that detect physical conditions and actuators that control physical devices
- D. By displaying websites on a monitor

Q17 (2 marks)

Why can manufacturers often produce embedded systems at lower cost than general-purpose computers with similar processing needs?

- A. Embedded systems require more expensive components
- B. General-purpose computers use cheaper components
- C. Cost is always identical regardless of system type
- D. Embedded systems can use simpler processors, less memory, and no complex operating system since they only need to perform specific tasks

Q18 (2 marks)

In a factory, a robot arm that welds car body panels contains which type of system?

- ○ **A.** A gaming computer
- ○ **B.** An embedded system that controls precise movements and welding operations
- ○ **C.** A general-purpose laptop
- ○ **D.** No computer system is involved

Q19 (2 marks)

What challenges exist when updating the software in an embedded system compared to updating a smartphone app?

- ○ **A.** Embedded systems may require physical access, specialised equipment, or careful testing since failures could render the device non-functional
- ○ **B.** Embedded system updates are always easier than app updates
- ○ **C.** Neither embedded systems nor smartphones can ever be updated
- ○ **D.** Updates work identically for all computer systems

Q20 (2 marks)

Why can embedded systems often be made physically smaller than general-purpose computers?

- ○ **A.** Embedded systems must always be larger than general computers

- ○ **B.** They need less memory, simpler processors, no full keyboard or display, and less cooling since they perform limited functions
- ○ **C.** Size is determined only by the colour of the case
- ○ **D.** General-purpose computers can be made any size while embedded systems cannot

Q21 (2 marks)

A hospital uses an infusion pump to deliver precise medication doses to patients. Why is this an embedded system?

- ○ **A.** Because it is located in a hospital
- ○ **B.** Because it connects to the internet
- ○ **C.** Because nurses use it
- ○ **D.** Because it contains a dedicated computer system programmed specifically to control medication delivery with no other functions

Q22 (2 marks)

What is true about embedded systems and network connectivity?

- ○ **A.** All embedded systems must be connected to the internet
- ○ **B.** Embedded systems never have any network capability
- ○ **C.** Some embedded systems have network connectivity for specific purposes while many operate independently with no network connection

○ **D.** Network connectivity automatically makes a device a general-purpose computer

Q23 (2 marks)

Why do general-purpose computers need complex operating systems while many embedded systems do not?

○ **A.** General-purpose computers are too simple for operating systems

○ **B.** General-purpose computers must manage multiple applications, user interfaces, file systems, hardware drivers, and security - complexity that dedicated embedded functions do not require

○ **C.** Operating systems make computers slower so are never used

○ **D.** Embedded systems always have more complex operating systems

Q24 (3 marks)

A modern car contains over 100 embedded systems. Analyse why manufacturers use separate embedded systems for different car functions rather than one central computer.

○ **A.** Multiple systems provide fault isolation (one failure does not crash everything), allow specialised optimisation for each function, enable independent development and testing, and simplify regulatory certification of safety-critical components

○ **B.** Having one central computer would be simpler and more reliable

○ **C.** Cost is the only consideration and separate systems are always cheaper

○ **D.** Cars do not contain any computer systems

Q25 (3 marks)

Internet of Things (IoT) devices combine embedded system characteristics with network connectivity. Analyse the unique challenges this creates.

○ **A.** IoT devices have no unique challenges compared to traditional embedded systems or general-purpose computers

○ **B.** IoT devices are identical to desktop computers in every way

○ **C.** Adding network connectivity automatically solves all embedded system limitations

○ **D.** IoT devices must balance embedded constraints (low power, limited resources, dedicated function) with security requirements (encryption, authentication, vulnerability patching) - creating tensions that neither traditional embedded nor desktop systems faced

Q26 (3 marks)

A medical device company is designing an insulin pump that will be implanted in diabetic patients. Analyse why embedded system characteristics are essential for this application.

○ **A.** General-purpose computers would be equally suitable for implanted medical devices

○ **B.** Embedded systems provide the reliability, miniature size, extremely low power consumption, instant response, and dedicated certified software required for a life-sustaining device implanted in the human body

○ **C.** Medical devices do not require computer systems

○ **D.** Cost is the only reason to choose an embedded system

Q27 (3 marks)

A smart home contains: (1) smart light bulbs, (2) a smart doorbell camera, (3) a central home automation hub. Analyse the embedded system characteristics of each and how they work together.

○ **A.** None of these devices contain embedded systems

○ **B.** Each device contains an embedded system optimised for its function: bulbs for lighting control, doorbell for video capture and motion detection, hub for integration and automation. They communicate over a network but remain dedicated to their specific purposes, with the hub coordinating actions across devices

○ **C.** Only general-purpose computers can control home automation

○ **D.** All three are identical devices with the same capabilities

Q28 (3 marks)

A company needs to process sensor data from a factory floor. Compare using a general-purpose computer running specialised software versus a dedicated embedded system for this task.

○ **A.** The embedded system offers higher reliability, real-time guarantees, and lower power; the general-purpose computer offers flexibility, easier updates, and familiar development tools. Choice depends on whether consistent real-time response or development flexibility is more critical

○ **B.** Both approaches are identical in every way

○ **C.** General-purpose computers cannot process sensor data

○ **D.** Embedded systems cannot connect to sensors

Q29 (3 marks)

Compare the software development lifecycle for an embedded system controlling an aircraft autopilot versus a smartphone app.

○ **A.** Both follow identical development processes with no differences

○ **B.** Neither requires any testing before release

○ **C.** Smartphone apps require more testing than aircraft systems

○ **D.** The autopilot requires extensive safety certification, formal verification methods, comprehensive testing, and years of development; the smartphone app uses agile development, rapid iteration, and quick updates. Safety-critical embedded systems demand rigour that consumer apps do not require

Q30 (3 marks)

As embedded systems become more powerful and internet-connected, some are beginning to resemble general-purpose computers. Analyse where the distinction between embedded and general-purpose systems remains meaningful.

○ **A.** The distinction has completely disappeared and is no longer relevant

○ **B.** Core distinctions remain: embedded systems are optimised for specific dedicated functions, prioritise reliability over flexibility, run manufacturer-controlled software, and must meet real-time or safety requirements. General-purpose systems prioritise user flexibility and run user-chosen software

○ **C.** Embedded systems will never become more powerful

○ **D.** General-purpose computers will be replaced entirely by embedded systems

Total: **55** marks • Time: **30 min**

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